

# Wireless Communication Numerical Companion

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

## Ch 7 · Multiple Access Numerical Problems & Solutions

Every problem below is a real PYQ, with its year tags and marks, except the four marked **PROBLEM**, which are tutorial or textbook problems added to cover a pattern the papers imply but never set. Solutions are worked in full: what was calculated, why, and which formula was used. **Nothing is skipped to save space.**

The four formulas everything here is built on

- **FDMA channels**  $N = \frac{B_t - 2B_{\text{guard}}}{B_c}$
- **TDMA channels**  $N = \frac{m(B_{\text{tot}} - 2B_{\text{guard}})}{B_c}$ ,  $m = \text{slots per carrier}$
- **TDMA frame efficiency**  $\eta_f = \left(1 - \frac{b_{OH}}{b_T}\right) \times 100\%$
- **Hadamard recursion**  $H_{2N} = \begin{bmatrix} H_N & H_N \\ H_N & -H_N \end{bmatrix}$ ,  $H_1 = [1]$

$B_t$  = total spectrum allocation,  $B_{\text{guard}}$  = guard band at *one* edge,  $B_c$  = channel (carrier) bandwidth,  $b_{OH}$  = overhead bits per frame,  $b_T$  = total bits per frame. **All three bandwidths must be in the same unit before you divide.** Convert everything to Hz, or everything to kHz, and never mix.

### Problem 1 Number of channels available in an FDMA system

**HOT 4/22** 80 Ba **3+2**, 79 Ch **4**, 77 Ch **4**, 78 Ch **4** · **same formula three times**, only the numbers change

**1.1**  $B_t = 25 \text{ MHz}$ , **guard** = 100 kHz, **channel** = 200 kHz (80 Ba)

**QUESTION AS PRINTED** 2080 Baishakh · Q7(a) · [3+2]

Explain the non-linear effect in FDMA. If the total spectrum allocation is 25 MHz, the guard band allocated at the edge of the spectrum is 100 KHz, and the channel bandwidth is 200 KHz. Find the number of channels available in an FDMA system.

The 3-mark theory half is answered in the chapter, §7.2. Only the 2-mark calculation is worked here.

- **Step 1: list the given data and put it all in one unit.** The formula divides bandwidths by each other, so a mixed unit silently gives an answer 1000 times wrong.
  - $B_t = 25 \text{ MHz} = 25 \times 10^6 \text{ Hz}$
  - $B_{\text{guard}} = 100 \text{ kHz} = 100 \times 10^3 = 0.1 \times 10^6 \text{ Hz}$
  - $B_c = 200 \text{ kHz} = 200 \times 10^3 = 0.2 \times 10^6 \text{ Hz}$
- **Step 2: subtract the guard bands.** **A guard band sits at each edge of the allocation**, at the bottom and at the top, so  $2B_{\text{guard}}$  is lost before any channel is cut. This factor of 2 is the single most common slip in this problem.

$$B_t - 2B_{\text{guard}} = 25 \times 10^6 - 2(100 \times 10^3) = 25,000,000 - 200,000 = 24,800,000 \text{ Hz}$$

- **Step 3: divide the usable spectrum by the channel bandwidth.**

$$N = \frac{B_t - 2B_{\text{guard}}}{B_c} = \frac{24,800,000}{200 \times 10^3} = \frac{24,800}{200} = 124$$

- **Step 4: state the answer as a whole number.**  $N$  counts channels, so it is an integer. Here the division is exact, and where it is not, **round down**: a fractional channel cannot be assigned.
- **Final answer:  $N = 124$  channels.**

**1.2**  $B_t = 12.8 \text{ MHz}$ , **guard** = 10 kHz, **channel** = 30 kHz (79 Ch, 77 Ch)

**QUESTION AS PRINTED** 2079 Chaitra · Q8(b) and 2077 Chaitra · Q10(b) · [4]

Calculate the number of radio channels available in the FDMA system with following data. A US analog mobile phone system is allocated 12.8 MHz for every simplex band. The total spectrum allocated is 12.8 MHz, the guard bandwidth is 10 KHz and the channel bandwidth is 30 KHz.

“fir” is the 2079 Chaitra paper’s own typo for “for”; 2077 Chaitra prints the same question correctly. The two papers are otherwise word for word identical.

- **Step 1: read the statement carefully.** It says 12.8 MHz twice: once as “allocated for every simplex band” and once as “the total spectrum allocated”. **These are the same number, not two numbers to add.**  $B_t = 12.8$  MHz is what goes into the formula.
  - The phrase “every simplex band” is telling you this is **one direction only**. A full duplex system would need a second 12.8 MHz for the reverse link, and the channel count would still be 12.8 MHz worth.
- **Step 2: convert to one unit.** Working in kHz keeps the numbers small:  $B_t = 12,800$  kHz,  $B_{\text{guard}} = 10$  kHz,  $B_c = 30$  kHz.
- **Step 3: remove both guard bands.**

$$B_t - 2B_{\text{guard}} = 12,800 - 2(10) = 12,800 - 20 = 12,780 \text{ kHz}$$

- **Step 4: divide by the channel bandwidth.**

$$N = \frac{12,780}{30} = 426$$

- **Final answer:  $N = 426$  radio channels.**

### 1.3 $B_t = 12.5$ MHz, guard = 10 kHz, channel = 30 kHz (78 Ch)

QUESTION AS PRINTED 2078 Chaitra · Q7(b) · [4]

If total bandwidth is 12.5 MHz, guard bandwidth is 10 KHz and channel bandwidth is 30 KHz, find the number of channels available in an FDMA system.

- **Step 1: same formula, in kHz.**  $B_t = 12,500$  kHz,  $B_{\text{guard}} = 10$  kHz,  $B_c = 30$  kHz.
- **Step 2: remove both guard bands.**

$$B_t - 2B_{\text{guard}} = 12,500 - 20 = 12,480 \text{ kHz}$$

- **Step 3: divide.**

$$N = \frac{12,480}{30} = 416$$

- **Final answer:  $N = 416$  channels.**
- **Worth knowing:** 416 is the real AMPS figure. In the US, each cellular carrier was allocated 416 channels per simplex band. **If your answer to this question is not 416, you have made an arithmetic slip.**

### 1.4 The same problem run backwards practice

PROBLEM not a PYQ · AS Sir, Tutorial VIII, Example 2 (alternate form)

In US AMPS, 416 channels are allocated to various cellular operators. The channel between them is 30 kHz with the guard band of 10 kHz. Calculate the spectrum allocation given to each operator.

- **Why this is worth doing:** the papers have only ever asked for  $N$ . The same formula rearranged asks for  $B_t$ , and **it is the obvious next variant** for a paper to set.
- **Step 1: rearrange the channel-count formula for  $B_t$ .** Starting from  $N = (B_t - 2B_{\text{guard}})/B_c$ , multiply both sides by  $B_c$  and add  $2B_{\text{guard}}$ :

$$B_t = NB_c + 2B_{\text{guard}}$$

- **Step 2: substitute.**

$$B_t = 416 \times (30 \times 10^3) + 2(10 \times 10^3)$$

- **Step 3: evaluate each term.**
  - Channels:  $416 \times 30,000 = 12,480,000 \text{ Hz} = 12.48 \text{ MHz}$ .
  - Guard bands:  $2 \times 10,000 = 20,000 \text{ Hz} = 0.02 \text{ MHz}$ .
- **Step 4: add.**

$$B_t = 12,480,000 + 20,000 = 12,500,000 \text{ Hz}$$

- **Final answer:  $B_t = 12.5$  MHz for each simplex band.**
- **Check:** this is exactly the input of Problem 1.3, which returned 416. The two problems are inverses of each other, so each one checks the other.

## Problem 2 Simultaneously transmitting users in an N-TDMA cluster

PIN 1/22 80 Ch 4

QUESTION AS PRINTED 2080 Chaitra · Q8(b) · [4]

In the N-TDMA system, it uses one way bandwidth of 25 MHz for the forward (or reverse) channel. The system bandwidth is divided into radio channels of 30 kHz. Guard bands with  $B_g = 20$  kHz are used. Calculate number of simultaneously transmitting users that can be accommodated in each cluster is.

The trailing “is” is the paper’s own slip. “Cluster” here means the whole system bandwidth, since in a cluster every channel is used exactly once.

- **Step 1: understand what is being counted.** A **cluster** is the group of cells over which the entire allocation is used once, with no repetition. So the number of simultaneously transmitting users per cluster is simply **the number of channels the whole allocation supports**.
  - Frequency reuse means the *next* cluster reuses the same channels, so the answer is per cluster, not per system.
  - **No cluster size  $N$  is needed**, and none is given. If the question had asked for users *per cell*, you would then divide by  $N$ .
- **Step 2: note which formula applies.** The statement gives no number of time slots per carrier, so take  $m = 1$  and the TDMA formula collapses to the FDMA one:

$$N_u = \frac{m(B_s - 2B_g)}{B_c} \xrightarrow{m=1} \frac{B_s - 2B_g}{B_c}$$

- **Step 3: list the data in kHz.**
  - $B_s = 25$  MHz = 25,000 kHz (one-way, so no halving is needed: the statement already says forward or reverse)
  - $B_g = 20$  kHz
  - $B_c = 30$  kHz
- **Step 4: remove both guard bands and divide.**

$$N_u = \frac{25,000 - 2(20)}{30} = \frac{25,000 - 40}{30} = \frac{24,960}{30} = 832$$

- **Final answer:  $N_u = 832$  simultaneously transmitting users per cluster.**
- **Sanity check:** the guard bands cost only  $40/25,000 = 0.16\%$  of the allocation, so the answer must land just under  $25,000/30 = 833.3$ . It does.

## Problem 3 TDMA frame efficiency in GSM

PIN 2/22 70 Bh 4, 74 Ma 2 · the numbers 156.25, 1250, 322 and 74.24 % are worth memorising outright

### 3.1 Find the frame efficiency (70 Bh)

QUESTION AS PRINTED 2070 Bhadra · Q8 · [2+6+4]

Define multiple access. What are the merits and demerits of Code Division Multiple Access? If a normal GSM time slot consists of 6 trailing bits, 8.25 guard bits, 26 training bits, and 2 traffic bursts of 58 bits of data, find the frame efficiency.

The first two parts are theory, answered in chapter §7.1 and §7.6. Only the 4-mark calculation is worked here.

- **Step 1: build one time slot, field by field.** Add every field the statement lists. **The 8.25 guard bits are not a typo:** GSM’s guard period really is a quarter bit longer than 8 bits, so the slot length comes out non-integer.

| Field                | Count         | Bits          | Overhead or payload?                       |
|----------------------|---------------|---------------|--------------------------------------------|
| Trailing (tail) bits | 6             | 6             | overhead, ramps the modulator              |
| Guard bits           | 8.25          | 8.25          | overhead, absorbs propagation delay spread |
| Training bits        | 26            | 26            | overhead, trains the equalizer             |
| Traffic bursts       | $2 \times 58$ | 116           | <b>payload</b> , the only useful bits      |
| <b>One time slot</b> |               | <b>156.25</b> |                                            |

- Arithmetic in full:  $6 + 8.25 + 26 + 2(58) = 6 + 8.25 + 26 + 116 = 156.25$  bits.
- **Step 2: build the frame.** **A GSM frame holds 8 time slots**, one per user. This number is not in the question, it is standard GSM knowledge you must supply.

$$b_T = 8 \times 156.25 = 1250 \text{ bits per frame}$$

- **Step 3: total the overhead over the whole frame.** Every one of the 8 slots carries its own tail, guard and training bits, so multiply each by 8:

$$b_{OH} = 8(6) + 8(8.25) + 8(26) = 48 + 66 + 208 = 322 \text{ bits}$$

- $8 \times 6 = 48$  (tail)
- $8 \times 8.25 = 66$  (guard)
- $8 \times 26 = 208$  (training)

- **Step 4: apply the frame efficiency formula.** Efficiency is the fraction of the frame that is *not* overhead:

$$\eta_f = \left(1 - \frac{b_{OH}}{b_T}\right) \times 100\% = \left(1 - \frac{322}{1250}\right) \times 100\%$$

- **Step 5: evaluate.**

$$\frac{322}{1250} = 0.2576 \rightarrow 1 - 0.2576 = 0.7424 \rightarrow \eta_f = 74.24\%$$

- **Final answer:  $\eta_f = 74.24\%$ .**
- **Cross-check the other way.** Payload per frame =  $8 \times 116 = 928$  bits, and  $928/1250 = 0.7424$ . **Same answer, which confirms that “overhead” was defined as everything except the two traffic bursts.**

### 3.2 Show that TDMA frame efficiency cannot reach 100 % in GSM (74 Ma)

QUESTION AS PRINTED 2074 Magh · Q8 · [6+2]

Explain briefly channel structure of GSM. Show that TDMA frame efficiency cannot reach 100% in GSM

The 6-mark GSM channel structure belongs to Ch 8. Only the 2-mark proof is here.

- **Step 1: write the definition as an inequality, not a number.**

$$\eta_f = \left(1 - \frac{b_{OH}}{b_T}\right) \times 100\%$$

so  $\eta_f = 100\%$  requires  $b_{OH} = 0$ .

- **Step 2: show  $b_{OH}$  can never be zero, field by field.** Each field is physically necessary, so none of them can be set to zero:
  - **Guard bits** absorb the difference in propagation delay between a mobile at the cell edge and one beside the base station. Remove them and **bursts from different mobiles overlap**, and both are lost.
  - **Training bits** give the adaptive equalizer a known sequence to estimate the channel from. The channel changes between bursts, so this must be repeated **in every burst**, not sent once.
  - **Tail bits** let the modulator ramp up and down inside the slot. Without them the burst edges splatter energy into adjacent channels.
  - **Preamble / synchronisation bits** let base station and subscriber identify each other and align in time.
- **Step 3: conclude with the actual number.** In GSM,  $b_{OH} = 322$  of  $b_T = 1250$  bits, i.e. **25.76 % of every frame is overhead**, giving  $\eta_f = 74.24\%$ . Since  $b_{OH} = 322 > 0$ ,  $\eta_f < 100\%$ .
- **Final answer:  $\eta_f = 74.24\% < 100\%$ , and it is bounded away from 100 % because the burst structure that makes TDMA work is itself made of non-payload bits.**
- **The general statement, worth one line: TDMA buys the ability to share a carrier by spending bits on framing.** FDMA transmits continuously and needs almost no framing, which is exactly why FDMA has lower overhead (chapter §7.10).

### 3.3 GSM slot and frame timing, and simultaneous users practice

PROBLEM not a PYQ · AS Sir, Tutorial VIII, Examples 3 and 4

Consider GSM, a TDMA/FDD system using 25 MHz for the forward link, broken into radio channels of 200 kHz. If 8 speech channels are supported on a single radio channel and no guard band is assumed, find the number of simultaneous users. If each frame consists of 8 time slots, each slot contains 156.25 bits, and data is transmitted at 270.833 kbps, find (a) the time duration of a bit, (b) of a slot, (c) of a frame, and (d) how long a user must wait between two transmissions.

- **Users: step 1, find the bandwidth one user really occupies.** A 200 kHz carrier is shared by 8 users, so **each user's share is  $200/8 = 25$  kHz**, even though the user briefly uses the whole 200 kHz.
- **Users: step 2, divide the allocation by that share.** No guard band is assumed, so nothing is subtracted:

$$N = \frac{25 \text{ MHz}}{200 \text{ kHz}/8} = \frac{25 \times 10^6}{25 \times 10^3} = 1000$$

**GSM can accommodate 1000 simultaneous users** in 25 MHz.

- (a) **Time duration of a bit.** The bit period is the reciprocal of the bit rate:

$$T_b = \frac{1}{270.833 \text{ kbps}} = \frac{1}{270,833} = 3.692 \times 10^{-6} \text{ s} = \mathbf{3.692 \mu s}$$

- (b) **Time duration of a slot.** A slot is 156.25 bits long, so multiply:

$$T_{\text{slot}} = 156.25 \times T_b = 156.25 \times 3.692 \mu s = 576.9 \mu s = \mathbf{0.577 \text{ ms}}$$

- (c) **Time duration of a frame.** A frame is 8 slots:

$$T_f = 8 \times T_{\text{slot}} = 8 \times 0.577 \text{ ms} = \mathbf{4.615 \text{ ms}}$$

- Cross-check through the bit count:  $1250 \text{ bits} \times 3.692 \mu s = 4.615 \text{ ms}$ . ✓
- So the frame rate is  $1/4.615 \text{ ms} = 216.7 \text{ frames per second}$ , which also equals  $270.833 \text{ kbps}/1250 \text{ bits}$ . ✓
- (d) **Wait between transmissions.** A user owns one slot in every frame, so after its burst ends it must wait for the *next* frame to come round: **4.615 ms**, one full frame time.
- **Why this matters:** 4.615 ms of buffering is why TDMA speech has more delay than FDMA speech, and the 0.577 ms burst is why the burst bit rate is 8 times the user's own rate, which is what forces the equalizer (chapter §7.3).

#### Problem 4 Construct the $8 \times 8$ Hadamard code

**HOT 3/22** 82 Bh **3**, 80 Bh **4**, 79 Ch **3+5**

**QUESTION AS PRINTED** 2079 Chaitra · Q7 **3+5** · also 2082 Bhadra Q6(a) **2+3**, 2080 Bhadra Q6(b) **4**

Write the steps to satisfy condition for Hadamard code. Construct 8 by 8 Hadamard code that satisfies above conditions.

2082 Bhadra prints “Define turbo coding. Construct 8 by 8 Hadamard code.” and 2080 Bhadra prints “Construct Hadamard matrix for  $H_8$ .” Same construction, three different mark splits.

##### Step 1 State the conditions the matrix must satisfy **3**

- It is a **square** matrix of order  $N$ , every entry  $+1$  or  $-1$ .
- **Every two distinct rows are orthogonal:** multiply them element by element, add up the products, and the result is 0.
- Equivalently  $H_N H_N^T = N I_N$ , where  $I_N$  is the  $N \times N$  identity.
- The order must be  $N = 1$ ,  $N = 2$ , or a **multiple of 4**. There is no  $3 \times 3$  or  $6 \times 6$  Hadamard matrix.
- The  $N$  rows are the  $N$  **codewords**, each  $N$  bits long, so the code carries  $k = \log_2 N$  information bits. For  $N = 8$ : **8 codewords, 8 bits each,  $k = 3$** .
- Minimum distance is  $d_{\min} = N/2 = 4$ , so it detects 3 errors and corrects  $\lfloor (4 - 1)/2 \rfloor = 1$  error.

##### Step 2 Start from $H_1$ and $H_2$

- The seed is the trivial  $1 \times 1$  matrix  $H_1 = [1]$ .
- Applying the recursion once gives the  $2 \times 2$  matrix everybody starts from:

$$H_2 = \begin{bmatrix} H_1 & H_1 \\ H_1 & -H_1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

- **Check it:** row 1 times row 2 =  $(1)(1) + (1)(-1) = 1 - 1 = 0$ . ✓ Orthogonal.

##### Step 3 Double it to $H_4$ using the Sylvester recursion

- The rule is  $H_{2N} = \begin{bmatrix} H_N & H_N \\ H_N & -H_N \end{bmatrix}$ : **copy the block three times and negate the bottom right one.**
- $-H_2$  means flip the sign of every entry:  $-H_2 = \begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$ .
- Assembling the four blocks:

$$H_4 = \begin{bmatrix} H_2 & H_2 \\ H_2 & -H_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$$

- **Check a pair:** row 2 times row 3 =  $(1)(1) + (-1)(1) + (1)(-1) + (-1)(-1) = 1 - 1 - 1 + 1 = 0$ . ✓

##### Step 4 Double once more to $H_8$ **5**

- Apply the same rule to  $H_4$ :  $H_8 = \begin{bmatrix} H_4 & H_4 \\ H_4 & -H_4 \end{bmatrix}$ .
- The top-left, top-right and bottom-left  $4 \times 4$  blocks are all  $H_4$  copied unchanged. The bottom-right block is  $H_4$  with every sign flipped.

$$H_8 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 & 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 & 1 & -1 & -1 & 1 \\ 1 & 1 & 1 & 1 & -1 & -1 & -1 & -1 \\ 1 & -1 & 1 & -1 & -1 & 1 & -1 & 1 \\ 1 & 1 & -1 & -1 & -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 & -1 & 1 & 1 & -1 \end{bmatrix}$$

### Step 5 Verify the orthogonality condition

- Pick any two distinct rows, multiply element by element, add. **Show at least one such check in the exam**, it is worth a mark.
- Row 2 against row 3:

$$\begin{aligned} (1)(1) + (-1)(1) + (1)(-1) + (-1)(-1) + (1)(1) + (-1)(1) + (1)(-1) + (-1)(-1) \\ = 1 - 1 - 1 + 1 + 1 - 1 - 1 + 1 = 0 \quad \checkmark \end{aligned}$$

- Row 1 against row 5:

$$1 + 1 + 1 + 1 - 1 - 1 - 1 - 1 = 0 \quad \checkmark$$

- Any row against itself:  $1 + 1 + 1 + 1 + 1 + 1 + 1 + 1 = 8 = N$ .  $\checkmark$  This is the second property CDMA decoding uses.

### Step 6 Convert to binary Walsh codes

- For transmission the  $\pm 1$  entries are mapped to bits: **+1  $\rightarrow$  0 and  $-1 \rightarrow$  1**.
- That turns the eight rows of  $H_8$  into the eight **Walsh codes**  $W_0 \dots W_7$ :

| Code  | Binary          | Ones | Code  | Binary          | Ones |
|-------|-----------------|------|-------|-----------------|------|
| $W_0$ | 0 0 0 0 0 0 0 0 | 0    | $W_4$ | 0 0 0 0 1 1 1 1 | 4    |
| $W_1$ | 0 1 0 1 0 1 0 1 | 4    | $W_5$ | 0 1 0 1 1 0 1 0 | 4    |
| $W_2$ | 0 0 1 1 0 0 1 1 | 4    | $W_6$ | 0 0 1 1 1 1 0 0 | 4    |
| $W_3$ | 0 1 1 0 0 1 1 0 | 4    | $W_7$ | 0 1 1 0 1 0 0 1 | 4    |

- **Check:** every codeword except the all-zero one has exactly  $N/2 = 4$  **ones**, which is the binary statement of  $d_{\min} = N/2$ .
- **Scaling up:** one more application of the recursion gives  $H_{64}$ , whose 64 rows are what IS-95 actually uses.  $W_0$ , the all-zero row, is its pilot channel. See chapter §7.7 for where these codes sit in the CDMA transmitter.

## Problem 5 CDMA encoding and decoding, worked with numbers

**HOT 3/22** 82 Bh **7**, 79 Ch **8**, 77 Ch **6** · the paper always says “with relevant example”, so an answer with no arithmetic loses most of the marks

**QUESTION AS PRINTED** 2082 Bhadra · Q7(a) **7** · 2079 Chaitra Q8(a) **8** · 2077 Chaitra Q10(a) **2+6**

Explain the implementation of CDMA with its encoding part and decoding part with relevant example.

2079 Chaitra prints “Illustrate Encoding and decoding mechanism of CDMA system in brief.” 2077 Chaitra prints “What are the major disadvantages of spread spectrum system? Explain the implementation of CDMA system with appropriate example.” The 2-mark disadvantages rider is answered in chapter §7.4 and §7.6.

### 5.1 Four stations, all transmitting

- **Step 1: write down the rules the arithmetic uses.** State these before calculating, they are worth marks on their own.
  1. Codes are **orthogonal**:  $C_i \cdot C_j = 0$  for  $i \neq j$ .
  2. A code times itself gives  $C_i \cdot C_i = N$ , the number of stations. Here  $N = 4$ .
  3. **Data mapping**: bit 1  $\rightarrow$  +1, bit 0  $\rightarrow$  -1, silent  $\rightarrow$  0.
  4. Multiplying a sequence by a number scales every element of it.
- **Step 2: set up the given data.** Four stations, data  $d_1 = 1$ ,  $d_2 = 0$ ,  $d_3 = 1$ ,  $d_4 = 1$ , and the four orthogonal chip sequences:

| Station | Chip sequence $C_i$      | Data bit $d_i$ | Mapped value |
|---------|--------------------------|----------------|--------------|
| ST1     | $C_1 = (+1, +1, +1, +1)$ | 1              | +1           |
| ST2     | $C_2 = (+1, -1, +1, -1)$ | 0              | -1           |
| ST3     | $C_3 = (+1, +1, -1, -1)$ | 1              | +1           |
| ST4     | $C_4 = (+1, -1, -1, +1)$ | 1              | +1           |

- **Step 3: confirm the codes really are orthogonal** before using them. Two spot checks are enough in the exam:
  - $C_1 \cdot C_2 = (1)(1) + (1)(-1) + (1)(1) + (1)(-1) = 1 - 1 + 1 - 1 = 0 \checkmark$
  - $C_3 \cdot C_4 = (1)(1) + (1)(-1) + (-1)(-1) + (-1)(1) = 1 - 1 + 1 - 1 = 0 \checkmark$
  - $C_1 \cdot C_1 = 1 + 1 + 1 + 1 = 4 = N \checkmark$
- **Step 4: encode. Each station multiplies its own code by its own data value.**
  - $d_1 C_1 = (+1)(+1, +1, +1, +1) = (+1, +1, +1, +1)$
  - $d_2 C_2 = (-1)(+1, -1, +1, -1) = (-1, +1, -1, +1)$
  - $d_3 C_3 = (+1)(+1, +1, -1, -1) = (+1, +1, -1, -1)$
  - $d_4 C_4 = (+1)(+1, -1, -1, +1) = (+1, -1, -1, +1)$
- **Step 5: all four transmit at once, and the channel adds them.** Sum the four sequences **element by element**, i.e. column by column:

| Contribution                         | chip 1    | chip 2    | chip 3    | chip 4    |
|--------------------------------------|-----------|-----------|-----------|-----------|
| $d_1 C_1$                            | +1        | +1        | +1        | +1        |
| $d_2 C_2$                            | -1        | +1        | -1        | +1        |
| $d_3 C_3$                            | +1        | +1        | -1        | -1        |
| $d_4 C_4$                            | +1        | -1        | -1        | +1        |
| <b>Common channel <math>S</math></b> | <b>+2</b> | <b>+2</b> | <b>-2</b> | <b>+2</b> |

- Column by column:  $1 - 1 + 1 + 1 = +2$ ,  $1 + 1 + 1 - 1 = +2$ ,  $1 - 1 - 1 - 1 = -2$ ,  $1 + 1 - 1 + 1 = +2$ .
- So the composite signal on the common channel is  $S = (+2, +2, -2, +2)$ .
- **Step 6: decode. Suppose ST3 wants ST2's data.** Multiply  $S$  by ST2's code element by element, add all four products, then divide by  $N = 4$ :

$$S \times C_2 = (+2)(+1), (+2)(-1), (-2)(+1), (+2)(-1) = (+2, -2, -2, -2)$$

$$\text{sum} = 2 - 2 - 2 - 2 = -4 \quad \rightarrow \quad \frac{-4}{4} = -1$$

- Applying the mapping backwards,  $-1$  means **bit 0**, which is exactly  $d_2$ .  $\checkmark$
- **Step 7: show the same receiver recovers every station.** This is the proof the scheme works, and it is where the last marks are:

| Decode with | $S \times C_i$ , element by element | Sum | $\div 4$ | Recovered bit               |
|-------------|-------------------------------------|-----|----------|-----------------------------|
| $C_1$       | $(+2, +2, -2, +2)$                  | +4  | +1       | <b>1</b> = $d_1 \checkmark$ |
| $C_2$       | $(+2, -2, -2, -2)$                  | -4  | -1       | <b>0</b> = $d_2 \checkmark$ |
| $C_3$       | $(+2, +2, +2, -2)$                  | +4  | +1       | <b>1</b> = $d_3 \checkmark$ |
| $C_4$       | $(+2, -2, +2, +2)$                  | +4  | +1       | <b>1</b> = $d_4 \checkmark$ |

- **Final answer: all four data bits (1, 0, 1, 1) are recovered from one shared 4-chip sequence**, with every station using the whole bandwidth for the whole time.
- **Why it works, in one line:** in  $S \times C_k$  every cross term  $d_i(C_i \cdot C_k)$  with  $i \neq k$  is zero by orthogonality, so only  $d_k(C_k \cdot C_k) = d_k N$  survives. Dividing by  $N$  returns  $d_k$ .

## 5.2 Three stations, one of them silent

**PROBLEM** not a PYQ · SST Ch 7, Figs 12.34 and 12.35 · the textbook version of the same question

Stations A, B and C are assigned codes  $C_A = (+1, -1, -1, +1)$ ,  $C_B = (+1, +1, +1, +1)$  and  $C_C = (+1, -1, +1, -1)$ . Station A wants to transmit a 0 bit, station B wants to remain silent, and station C wants to transmit a 1. Find the multiplexed CDMA signal, and show that a demultiplexer recovers all three states.

- **Why work this one too:** it is the version drawn in the lecture figures, and it adds the **silent station** case, which the four-station example does not cover. A paper asking "with relevant example" accepts either.
- **Step 1: map the three states.** Bit 0  $\rightarrow -1$ , **silent**  $\rightarrow 0$ , bit 1  $\rightarrow +1$ .

- A: bit 0  $\rightarrow -1$
- B: silent  $\rightarrow 0$
- C: bit 1  $\rightarrow +1$
- **Step 2: encode each station.**
  - $-1 \cdot C_A = (-1)(+1, -1, -1, +1) = (-1, +1, +1, -1)$
  - $0 \cdot C_B = (0)(+1, +1, +1, +1) = (0, 0, 0, 0)$  **a silent station contributes nothing at all**
  - $+1 \cdot C_C = (+1)(+1, -1, +1, -1) = (+1, -1, +1, -1)$
- **Step 3: add element by element to get the multiplexed signal.**
  - chip 1:  $-1 + 0 + 1 = 0$
  - chip 2:  $+1 + 0 - 1 = 0$
  - chip 3:  $+1 + 0 + 1 = +2$
  - chip 4:  $-1 + 0 - 1 = -2$

$$S = (0, 0, +2, -2)$$

- **Step 4: demultiplex each branch.** Multiply, add all bits, divide by  $N = 4$ :

| Branch   | $S \times C_i$ , element by element | Add all bits | $\div 4$ | Result          |
|----------|-------------------------------------|--------------|----------|-----------------|
| A, $C_A$ | (0, 0, -2, -2)                      | -4           | -1       | <b>bit 0</b> ✓  |
| B, $C_B$ | (0, 0, +2, -2)                      | 0            | 0        | <b>silent</b> ✓ |
| C, $C_C$ | (0, 0, +2, +2)                      | +4           | +1       | <b>bit 1</b> ✓  |

- **Final answer:**  $S = (0, 0, +2, -2)$ , and the decoder returns **bit 0, silent, bit 1**, matching what the three stations sent.
- **Note the three-way result.** The decoder does not just distinguish 1 from 0. It also reports *silence* as an exact zero, which is what lets CDMA gain capacity from **voice activity**: a silent user costs the system nothing.

### Problem 6 Capacity of an IS-95 CDMA cell practice

**No paper 2070–2082 has set this**, but it is the one calculation the CDMA theory implies and it is a standard tutorial problem. Included so the pattern is covered.

**PROBLEM** not a PYQ · AS Sir, Tutorial VIII, Example 8

In an IS-95 CDMA system, if  $W = 1.25$  MHz,  $R = 9600$  bps and  $N = 14$  users, (a) calculate  $E_b/N_0$ ; (b) when there is no voice activity, calculate it for omnidirectional antennas; (c) if voice activity =  $3/8$  and 3-sector antennas are used, calculate the total number of users per cell.

- **Step 1: find the processing gain.** It is the ratio of spread bandwidth to data rate:

$$\frac{W}{R} = \frac{1.25 \times 10^6}{9600} = 130.21$$

- **(a) Step 2: apply the single-cell CDMA capacity relation.** Each of the other  $N - 1$  users contributes an equal share of interference, so

$$\frac{E_b}{N_0} = \frac{W/R}{N-1} = \frac{130.21}{14-1} = \frac{130.21}{13} = 10.02$$

- In decibels,  $10 \log_{10}(10.02) = 10.01$  dB. **Final answer:**  $E_b/N_0 \approx 10$ , i.e. **about 10 dB**.
  - The linear value 10.02 and the dB value 10.01 coincide here only because  $10 \log_{10} 10 = 10$ . Do not assume that in other problems.
- **(b) With no voice activity and omnidirectional antennas, nothing changes.** Voice activity factor  $\alpha = 1$  means every user talks all the time, which is the assumption already built into part (a). An omnidirectional antenna gives no sectorisation gain. **So  $E_b/N_0$  stays at 10.02, about 10 dB.**
- **(c) Step 3: rearrange the relation for  $N$  and add the voice activity factor.** Solving  $E_b/N_0 = (W/R)/(N-1)$  for  $N$  gives  $N = 1 + (W/R)/(E_b/N_0)$ . A user who is silent a fraction of the time contributes interference only while talking, so the term is divided by  $\alpha$ :

$$N = 1 + \frac{1}{\alpha} \left[ \frac{W/R}{E_b/N_0} \right]$$

- **Step 4: substitute  $\alpha = 3/8$ , so  $1/\alpha = 8/3$ :**

$$N = 1 + \frac{8}{3} \left[ \frac{130.21}{10} \right] = 1 + \frac{8}{3}(13.021) = 1 + 34.72 = 35.72$$

i.e. about **35.7 users per sector**.

- **Step 5: apply the sectorisation gain.** A 3-sector antenna splits the cell into three, and each sector only hears the interference of its own users, so the cell carries three times the per-sector figure:

$$N_{\text{cell}} = 3 \times 35.72 = 107.2$$

- **Final answer: about 107 users per cell.**
- **Reading the result:** the raw processing gain alone would allow 14 users. Voice activity multiplies that by  $8/3$  and sectoring by 3, taking it to roughly 107. **This is exactly the “soft capacity” argument of chapter §7.6 in numbers:** CDMA capacity is set by the interference budget, not by a fixed count of channels.

### Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short    | Means                                   | Short    | Means                                      |
|----------|-----------------------------------------|----------|--------------------------------------------|
| BS       | base station                            | BTS      | base transceiver station                   |
| MS       | mobile station                          | MSC      | mobile switching centre                    |
| BSC      | base station controller                 | MSK      | minimum shift keying                       |
| CCI      | co-channel interference                 | ACI      | adjacent channel interference              |
| SIR, S/I | signal to interference ratio            | SNR, S/N | signal to noise ratio                      |
| GoS      | grade of service                        | MAHO     | mobile assisted handoff                    |
| FCA      | fixed channel assignment                | DCA      | dynamic channel assignment                 |
| LOS      | line of sight                           | NLOS     | non line of sight                          |
| FDMA     | frequency division multiple access      | TDMA     | time division multiple access              |
| CDMA     | code division multiple access           | OFDM     | orthogonal frequency division multiplexing |
| FDD      | frequency division duplex               | TDD      | time division duplex                       |
| GSM      | global system for mobile communications | AMPS     | advanced mobile phone system               |
| PCS      | personal communication system           | DAS      | distributed antenna system                 |
| BCCH     | broadcast control channel               | EIRP     | effective isotropic radiated power         |
| ERP      | effective radiated power                | RF       | radio frequency                            |
| Tx, TX   | transmitter                             | Rx, RX   | receiver                                   |

  

| Short   | Means                                       | Short | Means                                  |
|---------|---------------------------------------------|-------|----------------------------------------|
| ISI     | intersymbol interference                    | PDP   | power delay profile                    |
| DFE     | decision feedback equalizer                 | MLSE  | maximum likelihood sequence estimation |
| LMS     | least mean square                           | RLS   | recursive least squares                |
| MSE     | mean square error                           | MMSE  | minimum mean square error              |
| ZF      | zero forcing                                | MRC   | maximal ratio combining                |
| EGC     | equal gain combining                        | BER   | bit error rate                         |
| PCM     | pulse code modulation                       | DPCM  | differential PCM                       |
| ADPCM   | adaptive differential PCM                   | LPC   | linear predictive coder                |
| RPE-LTP | regular pulse excited, long term prediction | CELP  | code excited linear prediction         |
| RELTP   | residual excited linear prediction          | MPE   | multipulse excited                     |
| VSELP   | vector sum excited linear prediction        | ATC   | adaptive transform coding              |
| SBC     | sub-band coding                             | DCT   | discrete cosine transform              |
| FEC     | forward error correction                    | ARQ   | automatic repeat request               |
| BCH     | Bose–Chaudhuri–Hocquenghem code             | RS    | Reed–Solomon code                      |
| TCM     | trellis coded modulation                    | PN    | pseudo-noise                           |
| MIMO    | multiple input multiple output              | OFDMA | orthogonal FDM access                  |

| Short      | Means                                  | Short         | Means                                      |
|------------|----------------------------------------|---------------|--------------------------------------------|
| QAM        | quadrature amplitude modulation        | PSK           | phase shift keying                         |
| GMSK       | Gaussian minimum shift keying          | IMT           | international mobile telecommunications    |
| ASK        | amplitude shift keying                 | FSK           | frequency shift keying                     |
| BPSK       | binary phase shift keying              | QPSK          | quadrature phase shift keying              |
| OQPSK      | offset (staggered) QPSK                | DPSK, DBPSK   | differential (binary) PSK                  |
| CPFSK      | continuous phase FSK                   | MFSK, MPSK    | M-ary FSK, M-ary PSK                       |
| DSSS       | direct sequence spread spectrum        | FHSS          | frequency hopped spread spectrum           |
| SS, SSM    | spread spectrum (modulation)           | PG            | processing gain                            |
| PAPR       | peak-to-average power ratio            | CP            | cyclic prefix                              |
| FFT, IFFT  | (inverse) fast Fourier transform       | ICI           | inter-carrier interference                 |
| PSD        | power spectral density                 | CFO, SFO      | carrier, sampling frequency offset         |
| NRZ, RZ    | non-return, return to zero             | AMI           | alternate mark inversion                   |
| AM, FM, PM | amplitude, frequency, phase modulation | DSB, SSB, VSB | double, single, vestigial sideband         |
| S/P, P/S   | serial-parallel converter              | D/A, A/D      | digital-analog converter                   |
| LPF, BPF   | low pass, band pass filter             | ISM           | industrial, scientific, medical band       |
| SC-FDMA    | single carrier FDMA                    | <i>BT</i>     | GMSK filter bandwidth $\times$ symbol time |

  

| Short     | Means                                                  | Short    | Means                                               |
|-----------|--------------------------------------------------------|----------|-----------------------------------------------------|
| PSTN      | public switched telephone network                      | VoIP     | voice over IP                                       |
| FVC, RVC  | forward, reverse voice channel                         | FCC, RCC | forward, reverse control channel                    |
| SSMA      | spread spectrum multiple access                        | SDMA     | space division multiple access                      |
| FHMA      | frequency hopped multiple access                       | DSMA     | direct sequence multiple access (= CDMA)            |
| FCDMA     | hybrid FDMA/CDMA                                       | DS/FHMA  | hybrid direct sequence / frequency hopped           |
| TCDMA     | time division CDMA                                     | TDFH     | time division frequency hopping                     |
| IM        | intermodulation (product)                              | MU-MIMO  | multi-user MIMO                                     |
| WLAN      | wireless local area network                            | WMAN     | wireless metropolitan area network                  |
| WPAN      | wireless personal area network                         | AP       | access point                                        |
| BSS, IBSS | basic service set, independent (ad hoc) BSS            | ESS      | extended service set                                |
| CSMA/CA   | carrier sense multiple access with collision avoidance | CSMA/CD  | ... with collision detection                        |
| DCF       | distributed coordination function                      | PCF      | point coordination function                         |
| RTS/CTS   | request to send / clear to send                        | CCK      | complementary code keying                           |
| WEP, WPA  | wired equivalent privacy, WiFi protected access        | U-NII    | unlicensed national information infrastructure band |
| IoT       | internet of things                                     | mmWave   | millimetre wave                                     |
| eMBB      | enhanced mobile broadband                              | URLLC    | ultra-reliable low latency comm.                    |

| Short       | Means                                     | Short            | Means                                    |
|-------------|-------------------------------------------|------------------|------------------------------------------|
| mMTC        | massive machine type comm.                | NOMA             | non-orthogonal multiple access           |
| D2D         | device to device                          | C-RAN            | cloud radio access network               |
| PCN         | personal communication network            | SMS              | short message service                    |
| attr(s)     | attribute(s)                              | no.              | number                                   |
| freq        | frequency                                 | info             | information                              |
| ops         | operations                                | config           | configuration                            |
| rm          | remove                                    | vs, Vs           | versus                                   |
| w/, w/o     | with, without                             | Eg               | example / for example                    |
| +Fig        | draw a figure                             | (I)              | importance / need / significance         |
| →           | leads to / therefore                      | +                | and                                      |
| BSS         | base station subsystem                    | NSS              | network and switching subsystem          |
| OSS         | operation support subsystem               | OMC              | operation and maintenance centre         |
| HLR         | home location register                    | VLR              | visitor location register                |
| AuC         | authentication centre                     | EIR              | equipment identity register              |
| CEIR        | central EIR                               | ME               | mobile equipment                         |
| SIM         | subscriber identity module                | IMSI             | international mobile subscriber identity |
| Short       | Means                                     | Short            | Means                                    |
| IMEI        | international mobile equipment identity   | TMSI             | temporary mobile subscriber identity     |
| Um, Abis, A | the three GSM interfaces                  | SS7              | signalling system no. 7                  |
| ARFCN       | absolute radio frequency channel number   | TN, TS           | time slot number, time slot              |
| TCH         | traffic channel                           | TCH/FS, TCH/HS   | full rate, half rate speech TCH          |
| CCH         | control channel                           | CCCH             | common control channel                   |
| DCCH        | dedicated control channel                 | SDCCH            | stand-alone dedicated control channel    |
| SACCH       | slow associated control channel           | FACCH            | fast associated control channel          |
| FCCH        | frequency correction channel              | SCH              | synchronisation channel                  |
| PCH         | paging channel                            | RACH             | random access channel                    |
| AGCH        | access grant channel                      | TB, GP, SF       | tail bits, guard period, stealing flag   |
| VAD         | voice activity detector                   | DTX              | discontinuous transmission               |
| DCS 1800    | digital cellular system, the 1800 MHz GSM | CRC              | cyclic redundancy check                  |
| ESN         | electronic serial number                  | PCB              | power control bit                        |
| Mcps, Ksps  | mega-chips, kilo-symbols per second       | PRAT             | paging channel rate field                |
| EVRC        | enhanced variable rate coder              | PDSN             | packet data serving node                 |
| PCF (CDMA)  | packet control function                   | RNC              | radio network controller                 |
| Short       | Means                                     | Short            | Means                                    |
| EPS, EPC    | evolved packet system, core               | eNodeB           | LTE base station                         |
| MME         | mobility management entity                | S-GW, P-GW       | serving, packet data network gateway     |
| HSS         | home subscriber server                    | PCRF             | policy and charging rules function       |
| IMS, VoLTE  | IP multimedia subsystem, voice over LTE   | CoMP             | coordinated multipoint                   |
| ITU-R       | ITU radiocommunication sector             | WRC              | world radiocommunication conference      |
| NTA         | Nepal Telecommunications Authority        | UGS, rtPS, nrtPS | WiMAX QoS service classes                |

**Standard and system names** (proper names, kept as printed): **AMPS** advanced mobile phone system · **ETACS** extended total access communication system · **NMT** Nordic mobile telephone · **NTT** Nippon Telegraph and Telephone · **IMTS** improved mobile telephone service · **PDC** Pacific digital cellular · **IS-95** cdmaOne · **IS-136** digital AMPS · **HSCSD** high speed circuit switched data · **GPRS** general packet radio service · **EDGE** enhanced data rates for GSM evolution · **W-CDMA** wideband CDMA · **UMTS** universal mobile telecommunications system · **cdma2000 1xRTT / 1xEV-DO / 1xEV-DV** the cdma2000 3G family · **LTE, LTE-A** long term evolution (advanced) · **WiMAX** IEEE 802.16 · **WiFi** IEEE 802.11 · **DECT** digital enhanced cordless telecommunications · **CT2** cordless telephone 2 · **PHS** personal handyphone system · **USDC** US digital cellular (IS-136) · **JDC** Japanese digital cellular (later PDC) · **Bluetooth** IEEE 802.15.1 · **Zigbee** IEEE 802.15.4 · **PACS** personal access communications system · **POCSAG, FLEX** paging protocols · **CDPD** cellular digital packet

data · **Mobitex** a dedicated packet data system · **TETRA** terrestrial trunked radio · **DAB** digital audio broadcasting · **DVB-T** digital video broadcasting, terrestrial · **ADSL** asymmetric digital subscriber line · **5G NR** 5G new radio.

**Three clashes to watch:** **FCC** is the *forward control channel* throughout these notes; the US regulator is always written out as the Federal Communications Commission, and **FCCH** is GSM's frequency correction channel, a different thing again. **BCH** is the Bose–Chaudhuri–Hocquenghem block code in **Ch 6** and GSM's *broadcast channel* in **Ch 8**. **SCH** is GSM's synchronisation channel, never a sub-channel.

**Exam-paper month codes** (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

**Bold** year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

**Tier chips** over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth  $\geq 6$  marks. **Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.